ALEXANDER G. LUCACI, PH.D.

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Computational evolutionary biologist developing scalable methods to study how microbial communities adapt across hosts, environments, and populations. My work integrates molecular evolution, selection inference, and genomic surveillance to identify evolutionary constraints, characterize complex microbiomes, and anticipate emerging biological change across global urban environments.

EDUCATION

Ph.D. in Bioinformatics, Temple University 2023
Dissertation: The Role of Complex Evolutionary Dynamics in Molecular Sequence Analysis

M.S. in Biology, New York University 2018

B.S. in Biochemistry, Stony Brook University 2011

RESEARCH APPOINTMENTS

Weill Cornell Medical College July 2023 to Present
Postdoctoral Associate *Advisor: Dr. Christopher E. Mason*

- Conduct interdisciplinary research at the interface of evolutionary biology, metagenomics, and viromics, with emphasis on the NIH Human Virome Program (HVP) and the MetaSUB Consortium. Apply molecular sequence analysis and evolutionary theory to large-scale genomic and metatranscriptomic datasets to characterize urban microbial and viral ecosystems.
- Develop scalable computational frameworks for selection analysis, urban virome surveillance, and real-time biological threat detection, contributing to global efforts in city-scale microbiome monitoring and predictive evolutionary modeling.

Temple University Spring 2020 to Spring 2023
Graduate Research Assistant *Advisor: Dr. Sergei L. Kosakovsky Pond*

- Designed and implemented computational methods in HyPhy to quantify the impact of multinucleotide substitutions (MNS) on molecular adaptation, contributing to the BUSTED+S+MH and FMM frameworks that explicitly model clustered mutational processes and improve the robustness of episodic and gene-wide dN/dS inference.

RESEARCH LEADERSHIP

NIH - Human Virome Program January 2025 to Present
Co-Chair: Bioinformatics Working Group

Coordinate bioinformatics strategy for the Human Virome Program initiative, developing standardized pipelines for viromics analysis and scalable methods for viral discovery and host profiling. Drive cross-disciplinary collaboration, multi-omic integration, and reproducible, open viral bioinformatics infrastructure.

The MetaSUB International Consortium June 2024 to Present
Executive Director / Director of Bioinformatics

Oversee global bioinformatics workflows for MetaSUB urban microbiome studies across 100+ cities, developing scalable pipelines for taxonomic, functional, AMR, and viral analyses. Coordinate international data harmonization, mentor trainees, and advance computational methods for environmental and public health microbiology.

PEER-REVIEWED PUBLICATIONS

1. Can Kurt, K., Katzin, L. F., Tamaddon, L., Arslan, A., **Lucaci, A. G.**, & Mason, C. E. (2026). Adaptive Response of Escherichia coli to Pexiganan: Insights from Genomic Analysis. Antibiotics, 15(9), 825. https://doi.org/10.3390/antibiotics15090825
2. Gao X, Sanui A, Rasmika Dewi DAP, **Lucaci AG**, Mason CE, Suzuki H. Shotgun metagenomic dataset of surface microbiomes at a train station in Shinagawa, Tokyo. BMC Genom Data. 2026 Jun 12;27(1):70. doi:10.1186/s12863-026-01451-5
3. Gedman GL, Pirovich KM, Oppenheimer J, Hyseni C, Cassatt-Johnstone M, **Lucaci AG**, et al. On the ancestry and evolution of the extinct dire wolf. Cell Genomics. 2026 Aug 12;6(8). doi:10.1016/j.xgen.2026.101306 PubMed PMID: 42508403.
4. Gao Z*, Wu J*, **Lucaci AG***, Ouyang J, Wang L, Ryon KA, Elhaik E, Probst AJ, Rod X, Velavan TP, Chasapi A, Ouzounis CA, Oliveira M, Dias-Neto E, Osuolale O, Poulsen M, Meleshko D, Bhattacharyya M, Ugalde JA, Tull A, Rubins KH, Sierra MA, Tierney BT, Prithiviraj B, Sharma NK, Munteanu V, Mangul S, Kurt KC, Ushio M, Mazur-Panasiuk N, Kopera K, Marszaek K, Kowalski M, Toscan RB, Branicki W, Pyr K, abaj PP, Subramanian B, Frolova A, Burkhart JG, Deng Y, Udekwu KI, Schriml LM, Hazrin-Chong NH, Suzuki H, Lee PKH, Camargo AP, Kyrpides NC, Liu D, Wang LF, Mason CE, Shi T; International MetaSUB Consortium. Diversity and distinctive characteristics of the global RNA virome in urban and peri-urban environments. Nat Commun. 2026 Jul 16;17(1):6397. doi: 10.1038/s41467-026-73605-z. PMID: 42463489; PMCID: PMC13377081. * Co-First authors
5. Lei H, Du S, Li C, Yung L, Wang P, Leung LY, Graham CA, Yen HL, Li Y, **Lucaci AG**, Mason CE, Lee PKH. Sustained Chlorination of Hospital Surfaces Restructures the Microbiome and Virome and Diversifies Resistance Genes. Environ Sci Technol. 2026 Jun 16;60(23):16514-16525. doi: 10.1021/acs.est.6c01505. Epub 2026 Jun 4. PMID: 42237904.
6. Gao, X., Sanui, A., Rasmika Dewi, D. A. P., **Lucaci, A. G.**, Mason, C. E., & Suzuki, H. (2026). Shotgun metagenomic dataset of surface microbiomes at a train station in Shinagawa, Tokyo. BMC Genomic Data. https://doi.org/10.1186/s12863-026-01451-5
7. Lei, H., Du, S., Li, C., Yung, L., Wang, P., Leung, L. Y., Graham, C. A., Yen, H.-L., Li, Y., **Lucaci, A. G.**, Mason, C. E., & Lee, P. K. H. (2026). Sustained Chlorination of Hospital Surfaces Restructures the Microbiome and Virome and Diversifies Resistance Genes. Environmental Science & Technology. https://doi.org/10.1021/acs.est.6c01505
8. **Lucaci, AG**, Shaikh, H., Chong, L. C., Tahzima, R., Forgia, M., Mansour, K., Sakaguchi, S., Nakagawa, S., Hou, X., Demina, T., Raj Jayaraj Mallika, F., Kupczok, A., Lytras, S., Debat, H., Charon, J., Urzo, M., Raco, M., Kim, R., Rivero, R., Olendraite, I. (2026). Consensus statement from the second RdRp Summit: Towards a unified framework for RNA virus biology. Peer Community Journal, 6. https://doi.org/10.24072/pcjournal.727
9. **Lucaci AG**, Pond SLK. (2026). AOC: A Snakemake workflow for the characterization of natural selection in protein-coding genes. Journal of Open Source Software, 11(121), 9872, https://doi.org/10.21105/joss.09872
10. Sharma, G., Munteanu, V., Ghiasi, N. M., Mahanta, U., Banerjee, J., Varma, S., Foschini, L., Ellrott, K., Mutlu, O., Ciorb, D., Ophoff, R. A., Bostan, V., Moore, J. H., Sousoni, D., Krishnan, A., **Lucaci, AG**, Tull, A., Mason, C. E., Dimian, M., Mangul, S. (2026). Towards a decentralized future for open-science databases. Nature Genetics, 15. https://doi.org/10.1038/s41588-026-02606-x
11. Mak L, Tierney B, Wei W, Ronkowski C, Toscan RB, Turhan B, Toomey M, Andrade-Martnez JS, Fu C, **Lucaci AG**, Solano AHB, Setubal JC, Henriksen JR, Zimmerman S, Kopbayeva M, Noyvert A, Iwan Z, Kar S, Nakazawa N, Meleshko D, Horyslavets D, Kantsypa V, Frolova A, Kahles A, Danko D, Elhaik E, Labaj P, Mangul S; International MetaSUB Consortium; Mason CE, Hajirasouliha I. CAMP: a modular metagenomics analysis system for integrated multistep data

exploration. NAR Genom Bioinform. 2026 Jan 16;8(1):lqaf172. doi: 10.1093/nargab/lqaf172. PMID: 41551931; PMCID: PMC12809600.

12. Zhu K, Amirali A, Auch B, Babler KM, Biswas P, Bowie K, Choudhary S, Currall BB, Grills GS, Healy HG, Liachko I, **Lucaci AG**, Mason CE, Sharkey M, Shigeno Risse-Adams O, Shukla BS, Sisson Z, Stevenson M, Williams SL, Zulli A, Peccia J, Solo-Gabriele HM. Proof-of-concept of host attribution of antimicrobial resistance genes using wastewater Hi-C metagenome sequencing. J Water Health. 2026 Feb;24(2):148-159. doi: 10.2166/wh.2025.162. Epub 2025 Dec 29. PMID: 41764388.

13. Munteanu V, Saldana MA, Dreifuss D, Ouyang WO, Ferdous J, Mohebbi F, Roseberry JS, Ciorba D, Bostan V, Gordeev V, Drabcinski N, Su JM, Kasianchuk N, Sharma NK, Knyazev S, Amann E, Lobiuc A, Covasa M, Crandall KA, Wu NC, Mason CE, Tierney BT, **Lucaci AG**, Ophoff RA, Gibas C, Rzymski P, Skums P, Solo-Gabriele H, Niko B, Zelikovsky A, Hlzer M, Smith A, Mangul S. SARS-CoV-2 wastewater genomic surveillance: approaches, challenges, and opportunities. Genome Biol. 2026 Jan 12;27(1):1. doi: 10.1186/s13059-025-03927-6. PMID: 41526978; PMCID: PMC12794521.

14. Lei H, Du S, Tong X, Chan WL, Leung MHY, Bifot KO, Bezdan D, Butler DJ, Danko DC, Green DC, Hernandez MT, Kelly FJ, **Lucaci AG**, Meydan C, Nieto-Caballero M, Ryon K, Tierney B, Udekwu KI, Young BG, Mason CE, Dybwad M, Lee PKH. Global biogeography of airborne viruses in public transit systems and their host interactions. Microbiome. 2025 Aug 29;13(1):193. doi: 10.1186/s40168-025-02173-z. PMID: 40883783; PMCID: PMC12395665.

15. Amirali A, Sharkey ME, Choudhary S, Babler KM, Beaver CC, Biswas P, Bowie KR, Burke T, Currall BB, Grills GS, Healy HG, **Lucaci AG**, Mason CE, McGuire M, Ramos R, Ruedaflores M, Solle NS, Schrer SC, Shukla BS, Stevenson M, Vidovi D, Williams SL, Zhu K, Zulli A, Peccia J, Solo-Gabriele HM. Long term assessment of SARS-CoV-2 in wastewater and the transition to evaluate additional viral targets. Sci Total Environ. 2025 Sep 15;995:180096. doi: 10.1016/j.scitotenv.2025.180096. Epub 2025 Jul 24. PMID: 40712541; PMCID: PMC12490206.

16. Selberg, A., Clark, N. L., Sackton, T. B., Muse, S. V., **Lucaci, AG**, Weaver, S., Nekrutenko, A., Chikina, M., & Pond, S. L. K. (2025). Minus the Error: Testing for Positive Selection in the Presence of Residual Alignment Errors. eLife, 14. https://doi.org/10.7554/eLife.106921.1

17. Melendez-Vazquez F, **Lucaci AG**, Selberg A, Clavel J, Rincon-Sandoval M, Santaquiteria A, White WT, Drabeck D, Carnevale G, Duarte-Ribeiro E, Miya M, Westneat MW, Baldwin CC, Hughes LC, Ort G, Kosakovsky Pond SL, Betancur-R R, Arcila D. Ecological interactions and genomic innovation fueled the evolution of ray-finned fish endothermy. Sci Adv. 2025 Jun 27;11(26):eads8488. doi: 10.1126/sciadv.ads8488. Epub 2025 Jun 25. PMID: 40561012; PMCID: PMC12190012.

18. Gordeev V, Hlzer M, Desir D, Goraichuk IV, Knyazev S, Solo-Gabriele H, Skums P, Karthikeyan S, Evans A, Agrawal S, **Lucaci AG**, Mason CE, Su JM, Gibas C, Nagarajan N, Peres da Silva R, Drabcinski N, Munteanu V, Zhan L, Rubin J, Wu NC, Trister A, Ciorba D, Bostan V, Lobiuc A, Covasa M, Ophoff RA, Zelikovsky A, Dimian M, Mangul S. Leveraging wastewater sequencing to strengthen global public health surveillance. BMC Glob Public Health. 2025 Mar 21;3(1):23. doi: 10.1186/s44263-025-00138-w. PMID: 40119458; PMCID: PMC11927347.

19. Zlenko OB, Kit MY, Bondarenko O, Gincu AE, **Lucaci AG**. Spatially-distributed viral evolutionary analysis of Influenza A Virus (IAV). *Biopolym Cell.* 2024;40(3):217.

20. Tierney BT, Foox J, Ryon KA, Butler D, Damle N, Young BG, Mozsary C, Babler KM, Yin X, Carattini Y, Andrews D, **Lucaci AG**, Solle NS, Kumar N, Shukla B, Vidovi D, Currall B, Williams SL, Schrer SC, Stevenson M, Amirali A, Beaver CC, Kobetz E, Boone MM, Reding B, Laine J, Comerford S, Lamar WE, Tallon JJ Jr, Wain Hirschberg J, Proszynski J, Al Ghalith G, Can Kurt K, Sharkey ME, Church GM, Grills GS, Solo-Gabriele HM, Mason CE. Towards geospatially-resolved public-health surveillance via wastewater sequencing. Nat Commun. 2024

Sep 27;15(1):8386. doi: 10.1038/s41467-024-52427-x. PMID: 39333485; PMCID: PMC11436780.

21. Overbey EG, Kim J, Tierney BT, Park J, Houerbi N, **Lucaci AG**, Garcia Medina S, Damle N, Najjar D, Grigorev K, Afshin EE, Ryon KA, Sienkiewicz K, Patras L, Klotz R, Ortiz V, MacKay M, Schweickart A, Chin CR, Sierra MA, Valenzuela MF, Dantas E, Nelson TM, Cekanaviciute E, Deards G, Foox J, Narayanan SA, Schmidt CM, Schmidt MA, Schmidt JC, Mullane S, Tigchelaar SS, Levitte S, Westover C, Bhattacharya C, Lucotti S, Wain Hirschberg J, Proszynski J, Burke M, Kleinman AS, Butler DJ, Loy C, Mzava O, Lenz J, Paul D, Mozsary C, Sanders LM, Taylor LE, Patel CO, Khan SA, Suhail Mohamad M, Byhaqui SGA, Aslam B, Gajadhar AS, Williamson L, Tandel P, Yang Q, Chu J, Benz RW, Siddiqui A, Hornburg D, Blease K, Moreno J, Boddicker A, Zhao J, Lajoie B, Scott RT, Gilbert RR, Lai Polo SH, Altomare A, Kruglyak S, Levy S, Ariyapala I, Beer J, Zhang B, Hudson BM, Rininger A, Church SE, Beheshti A, Church GM, Smith SM, Crucian BE, Zwart SR, Matei I, Lyden DC, Garrett-Bakelman F, Krumsiek J, Chen Q, Miller D, Shuga J, Williams S, Nemec C, Trudel G, Pelchat M, Laneuville O, De Vlaminck I, Gross S, Bolton KL, Bailey SM, Granstein R, Furman D, Melnick AM, Costes SV, Shirah B, Yu M, Menon AS, Mateus J, Meydan C, Mason CE. The Space Omics and Medical Atlas (SOMA) and international astronaut biobank. Nature. 2024 Aug;632(8027):1145-1154. doi: 10.1038/s41586-024-07639-y. Epub 2024 Jun 11. PMID: 38862028; PMCID: PMC11357981.

22. Mason CE, Green J, Adamopoulos KI, Afshin EE, Baechle JJ, Basner M, Bailey SM, Bielski L, Borg J, Borg J, Broddrick JT, Burke M, Caicedo A, Castaeda V, Chatterjee S, Chin CR, Church G, Costes SV, De Vlaminck I, Desai RI, Dhir R, Diaz JE, Etlin SM, Feinstein Z, Furman D, Garcia-Medina JS, Garrett-Bakelman F, Giacomello S, Gupta A, Hassanin A, Houerbi N, Irby I, Javorsky E, Jirak P, Jones CW, Kamal KY, Kangas BD, Karouia F, Kim J, Kim JH, Kleinman AS, Lam T, Lawler JM, Lee JA, Limoli CL, **Lucaci AG**, MacKay M, McDonald JT, Melnick AM, Meydan C, Mieczkowski J, Muratani M, Najjar D, Othman MA, Overbey EG, Paar V, Park J, Paul AM, Perdyan A, Proszynski J, Reynolds RJ, Ronca AE, Rubins K, Ryon KA, Sanders LM, Glowe PS, Shevde Y, Schmidt MA, Scott RT, Shirah B, Sienkiewicz K, Sierra MA, Siew K, Theriot CA, Tierney BT, Venkateswaran K, Hirschberg JW, Walsh SB, Walter C, Winer DA, Yu M, Zea L, Mateus J, Beheshti A. A second space age spanning omics, platforms and medicine across orbits. Nature. 2024 Aug;632(8027):995-1008. doi: 10.1038/s41586-024-07586-8. Epub 2024 Jun 11. PMID: 38862027; PMCID: PMC12366838.

23. **Lucaci AG**, Brew WE, Lamanna J, Selberg A, Carnevale V, Moore AR, Kosakovsky Pond SL. The evolution of mammalian Rem2: unraveling the impact of purifying selection and coevolution on protein function, and implications for human disorders. Front Bioinform. 2024 Jun 24;4:1381540. doi: 10.3389/fbinf.2024.1381540. PMID: 38978817; PMCID: PMC11228553.

24. Garcia-Medina JS, Sienkiewicz K, Narayanan SA, Overbey EG, Grigorev K, Ryon KA, Burke M, Proszynski J, Tierney B, Schmidt CM, Mencia-Trinchant N, Klotz R, Ortiz V, Foox J, Chin C, Najjar D, Matei I, Chan I, Cruchaga C, Kleinman A, Kim J, **Lucaci AG**, Loy C, Mzava O, De Vlaminck I, Singaraju A, Taylor LE, Schmidt JC, Schmidt MA, Blease K, Moreno J, Boddicker A, Zhao J, Lajoie B, Altomare A, Kruglyak S, Levy S, Yu M, Hassane DC, Bailey SM, Bolton K, Mateus J, Mason CE. Genome and clonal hematopoiesis stability contrasts with immune, cfDNA, mitochondrial, and telomere length changes during short duration spaceflight. Precis Clin Med. 2024 Apr 8;7(1):pbae007. doi: 10.1093/pcmedi/pbae007. PMID: 38634106; PMCID: PMC11022651.

25. **Lucaci AG**, Zehr JD, Enard D, Thornton JW, Kosakovsky Pond SL. Evolutionary Shortcuts via Multinucleotide Substitutions and Their Impact on Natural Selection Analyses. Mol Biol Evol. 2023 Jul 5;40(7):msad150. doi: 10.1093/molbev/msad150. PMID: 37395787; PMCID: PMC10336034.

26. Zehr JD, Kosakovsky Pond SL, Millet JK, Olarte-Castillo XA, **Lucaci AG**, Shank SD, Ceres KM, Choi A, Whittaker GR, Goodman LB, Stanhope MJ. Natural selection differences detected in key protein domains between non-pathogenic and pathogenic feline coronavirus phenotypes.

Virus Evol. 2023 Mar 15;9(1):vead019. doi: 10.1093/ve/vead019. PMID: 37038392; PMCID: PMC10082545.

27. Silva SR, Miranda VFO, Michael TP, Pachno BJ, Matos RG, Adamec L, Pond SLK, **Lucaci AG**, Pinheiro DG, Varani AM. The phylogenomics and evolutionary dynamics of the organellar genomes in carnivorous Utricularia and Genlisea species (Lentibulariaceae). Mol Phylogenet Evol. 2023 Apr;181:107711. doi: 10.1016/j.ympev.2023.107711. Epub 2023 Jan 21. PMID: 36693533.
28. Tegally H, Moir M, Everatt J, Giovanetti M, Scheepers C, Wilkinson E, Subramoney K, Makatini Z, Moyo S, Amoako DG, Baxter C, Althaus CL, Anyaneji UJ, Kekana D, Viana R, Giandhari J, Lessells RJ, Maponga T, Maruapula D, Choga W, Matshaba M, Mbulawa MB, Msomi N; **NGS-SA consortium**; Naidoo Y, Pillay S, Sanko TJ, San JE, Scott L, Singh L, Magini NA, Smith-Lawrence P, Stevens W, Dor G, Tshiabuila D, Wolter N, Preiser W, Treurnicht FK, Venter M, Chiloane G, McIntyre C, O'Toole A, Ruis C, Peacock TP, Roemer C, Kosakovsky Pond SL, Williamson C, Pybus OG, Bhiman JN, Glass A, Martin DP, Jackson B, Rambaut A, Laguda-Akingba O, Gaseitsiwe S, von Gottberg A, de Oliveira T. Emergence of SARS-CoV-2 Omicron lineages BA.4 and BA.5 in South Africa. Nat Med. 2022 Sep;28(9):1785-1790. doi: 10.1038/s41591-022-01911-2. Epub 2022 Jun 27. PMID: 35760080; PMCID: PMC9499863.
29. **Lucaci AG**, Notaras MJ, Kosakovsky Pond SL, Colak D. The evolution of BDNF is defined by strict purifying selection and prodomain spatial coevolution, but what does it mean for human brain disease? Transl Psychiatry. 2022 Jun 22;12(1):258. doi: 10.1038/s41398-022-02021-w. PMID: 35732627; PMCID: PMC9217794.
30. Benndorf R, Velazquez R, Zehr JD, Pond SLK, Martin JL, **Lucaci AG**. Human HspB1, HspB3, HspB5 and HspB8: Shaping these disease factors during vertebrate evolution. Cell Stress Chaperones. 2022 Jul;27(4):309-323. doi: 10.1007/s12192-022-01268-y. Epub 2022 Jun 9. PMID: 35678958; PMCID: PMC9346038.
31. Viana R, Moyo S, Amoako DG, Tegally H, Scheepers C, Althaus CL, Anyaneji UJ, Bester PA, Boni MF, Chand M, Choga WT, Colquhoun R, Davids M, Deforche K, Doolabh D, du Plessis L, Engelbrecht S, Everatt J, Giandhari J, Giovanetti M, Hardie D, Hill V, Hsiao NY, Iranzadeh A, Ismail A, Joseph C, Joseph R, Koopile L, Kosakovsky Pond SL, Kraemer MUG, Kuate-Lere L, Laguda-Akingba O, Lesetedi-Mafoko O, Lessells RJ, Lockman S, **Lucaci AG**, Maharaj A, Mahlangu B, Maponga T, Mahlakwane K, Makatini Z, Marais G, Maruapula D, Masupu K, Matshaba M, Mayaphi S, Mbhele N, Mbulawa MB, Mendes A, Mlisana K, Mnguni A, Mohale T, Moir M, Moruisi K, Mosepele M, Motsatsi G, Motswaledi MS, Mphoyakgosi T, Msomi N, Mwangi PN, Naidoo Y, Ntuli N, Nyaga M, Olubayo L, Pillay S, Radibe B, Ramphal Y, Ramphal U, San JE, Scott L, Shapiro R, Singh L, Smith-Lawrence P, Stevens W, Strydom A, Subramoney K, Tebeila N, Tshiabuila D, Tsui J, van Wyk S, Weaver S, Wibmer CK, Wilkinson E, Wolter N, Zarebski AE, Zuze B, Goedhals D, Preiser W, Treurnicht F, Venter M, Williamson C, Pybus OG, Bhiman J, Glass A, Martin DP, Rambaut A, Gaseitsiwe S, von Gottberg A, de Oliveira T. Rapid epidemic expansion of the SARS-CoV-2 Omicron variant in southern Africa. Nature. 2022 Mar;603(7902):679-686. doi: 10.1038/s41586-022-04411-y. Epub 2022 Jan 7. PMID: 35042229; PMCID: PMC8942855.
32. **Lucaci AG**, Zehr JD, Shank SD, Bouvier D, Ostrovsky A, Mei H, Nekrutenko A, Martin DP, Kosakovsky Pond SL. RASCL: Rapid Assessment of Selection in CLades through molecular sequence analysis. PLoS One. 2022 Nov 2;17(11):e0275623. doi: 10.1371/journal.pone.0275623. PMID: 36322581; PMCID: PMC9629619.
33. Martin DP, Lytras S, **Lucaci AG**, Maier W, Grning B, Shank SD, Weaver S, MacLean OA, Orton RJ, Lemey P, Boni MF, Tegally H, Harkins GW, Scheepers C, Bhiman JN, Everatt J, Amoako DG, San JE, Giandhari J, Sigal A; NGS-SA; Williamson C, Hsiao NY, von Gottberg A, De Klerk A, Shafer RW, Robertson DL, Wilkinson RJ, Sewell BT, Lessells R, Nekrutenko A, Greaney AJ, Starr TN, Bloom JD, Murrell B, Wilkinson E, Gupta RK, de Oliveira T, Kosakovsky Pond SL. Selection Analysis Identifies Clusters of Unusual Mutational Changes in Omicron Lineage

BA.1 That Likely Impact Spike Function. Mol Biol Evol. 2022 Apr 11;39(4):msac061. doi: 10.1093/molbev/msac061. PMID: 35325204; PMCID: PMC9037384.

34. Martin DP, Weaver S, Tegally H, San JE, Shank SD, Wilkinson E, **Lucaci AG**, Giandhari J, Naidoo S, Pillay Y, Singh L, Lessells RJ; NGS-SA; COVID-19 Genomics UK (COG-UK); Gupta RK, Wertheim JO, Nekturenko A, Murrell B, Harkins GW, Lemey P, MacLean OA, Robertson DL, de Oliveira T, Kosakovsky Pond SL. The emergence and ongoing convergent evolution of the SARS-CoV-2 N501Y lineages. Cell. 2021 Sep 30;184(20):5189-5200.e7. doi: 10.1016/j.cell.2021.09.003. Epub 2021 Sep 7. PMID: 34537136; PMCID: PMC8421097.
35. **Lucaci AG**, Wisotsky SR, Shank SD, Weaver S, Kosakovsky Pond SL. Extra base hits: Widespread empirical support for instantaneous multiple-nucleotide changes. PLoS One. 2021 Mar 12; 16(3):e0248337. doi: 10.1371/journal.pone.0248337. PMID: 33711070; PMCID: PMC7954308.
36. Yao C, Vanderpool KG, Delfiner M, Eddy V, **Lucaci AG**, Soto-Riveros C, Yasumura T, Rash JE, Pereda AE. Electrical synaptic transmission in developing zebrafish: properties and molecular composition of gap junctions at a central auditory synapse. J Neurophysiol. 2014 Nov 1;112(9):2102-13. doi: 10.1152/jn.00397.2014. Epub 2014 Jul 30. PMID: 25080573; PMCID: PMC4274921.

PREPRINTS

1. Pond SLK, Weaver S, Callan D, Zehr JD, **Lucaci AG**, Verdonk H, et al. HyphAeon: Attention on Evolution Across Deep Time Transforms Comparative Genomics [Internet]. bioRxiv; 2026 [cited 2026 Oct 5]. p. 2026.09.06.749597. Available from: https://www.biorxiv.org/content/10.64898/2026.09.06.749597v1 doi:10.64898/2026.09.06.749597
2. Ouyang, J., Wu, J., Gao, Z., Rod, X., Elhaik, E., Toscan, R., abaj, P. P., Udekwu, K. I., Jang, S., **Lucaci, A. G.**, Proszynski, J., Mason, C. E., Shi, T., & Consortium, T. I. M. (2026). A Global Atlas of the Pan-Urban DNA Virome Reveals Extensive Viral Diversity and Its Links to Ecology and Public Health (SSRN Scholarly Paper No. 6403392). Social Science Research Network. https://doi.org/10.2139/ssrn.6403392
3. Oleksyk TK, Yakymenko D, Boek S, Munteanu V, Pilch W, Comarova Z, et al. Leveraging a hybrid cross-disciplinary training model to accelerate global bioinformatics capacity [Internet]. bioRxiv; 2026 [cited 2026 Feb 15]. p. 2026.01.21.700760.
4. **Lucaci AG**, Judah BJ, Risse-Adams OS, Saygili EA, Rod X, et al. Harnessing the global urban microbiome for the future of cities [Internet]. Zenodo; 2026 [cited 2026 Feb 15].

BOOK CHAPTERS

Fundamentals of Space Medicine and Clinical Technology: Viral evolution in the cosmos
Synthesizes evolutionary dynamics of viruses across extreme and extraterrestrial environments. Highlights selection pressures and adaptation in constrained ecosystems using comparative genomics frameworks. Connects viromics and astrobiology to predictive models of biological discovery and planetary health. Positions viral evolution as a framework for understanding biological adaptation in novel and engineered environments.

- Trovao NS, **Lucaci AG**, Pradeep N, Sethi M, Bono LM, Singh RS, Clark KB, Zaksas V, Burke M, Caicedo A, Castaeda V, Zambrano K, Karim R, Theriot CA, Otkiran G, Neefs D, Isola G, Tartaglia G, Schotsaert M, Tamim S, Das S, Fasullo M, Ali N, Fargette D, Brereton NJB, Beheshti A, Guarnieri JW, Wurtele ES. *Viral evolution in the cosmos.* In: *Fundamentals of Space Medicine and Clinical Technology.* Academic Press; **2026**. pp. 67–96. doi: 10.1016/B978-0-443-32904-3.00009-1.

SOFTWARE, DATA RESOURCES, AND WORKFLOWS

1. **RASCL: Rapid Assessment of Selection in Clades** *Lead developer / first author*
   Computational workflow for rapid molecular selection analysis across clades, supporting scalable and reproducible evolutionary analysis of protein-coding sequences. Designed to enable rapid assessment of selection patterns across viral and comparative genomic datasets.
   Associated publication: **Lucaci AG**, Zehr JD, Shank SD, Bouvier D, Ostrovsky A, Mei H, Nekrutenko A, Martin DP, Kosakovsky Pond SL. RASCL: Rapid Assessment of Selection in CLades through molecular sequence analysis. *PLOS One*. 2022 Nov 2;17(11):e0275623. doi: 10.1371/journal.pone.0275623. PMID: 36322581; PMCID: PMC9629619.
2. **AOC: Analysis of Orthologous Collections** *Lead developer / first author*
   Application for organizing orthologous protein-coding sequence collections and characterizing natural selection across genes and taxa. Developed to support reproducible comparative molecular evolutionary analyses across large orthologous datasets.
   Associated preprint: **Lucaci AG**, Pond SLK. AOC: Analysis of Orthologous Collections – an application for the characterization of natural selection in protein-coding sequences. *ArXiv* [Preprint]. 2024 Jun 13:arXiv:2406.09522v1. PMID: 38947939; PMCID: PMC11213150.
3. **HyPhy-based molecular evolution methods** *Lead method developer / first author*
   Designed and implemented computational approaches for evaluating the impact of multinucleotide substitutions and clustered mutational processes on molecular adaptation inference. Contributed to model development and workflow implementation supporting robust episodic and gene-wide selection analyses, including BUSTED+S+MH and FMM-related frameworks.
   Associated publication: **Lucaci AG**, Zehr JD, Enard D, Thornton JW, Kosakovsky Pond SL. Evolutionary Shortcuts via Multinucleotide Substitutions and Their Impact on Natural Selection Analyses. *Molecular Biology and Evolution*. 2023 Jul 5;40(7):msad150. doi: 10.1093/molbev/msad150. PMID: 37395787; PMCID: PMC10336034.
4. **Reproducible training workflows for global bioinformatics education** *Instructor*
   Developed hands-on instructional workflows for molecular adaptation, metagenomics, sequence analysis, and reproducible computational biology through international workshops and training programs. Materials emphasize practical analysis of real biological datasets using accessible, reproducible computational methods.
   Associated preprint: Oleksyk TK, Yakymenko D, Boek S, Munteanu V, Pilch W, Comarova Z, et al. Leveraging a hybrid cross-disciplinary training model to accelerate global bioinformatics capacity [Internet]. *bioRxiv*; 2026 [cited 2026 Feb 15]. p. 2026.01.21.700760.
   Available from: https://www.biorxiv.org/content/10.64898/2026.01.21.700760v1

MENTORING

1. **Mentor trainees in bioinformatics, metagenomics, viromics, molecular evolution, project design, data analysis, scientific writing, and professional development.** *2021–Present*
   - **Oona Shingeo Risse-Adams** – Graduate Student
   - **Betty Judah** – Medical Student
   - **Nesara Dandiganahalli Lakshmiprasad** – Masters Student
   - **Ayeh Khorshidian** – Masters Student
   - **Gabriella Smyth** – High School Student

GRANTS, FELLOWSHIPS, AND AWARDS

- **2026** NIH Loan Repayment Program (LRP) Award Renewal, *National Cancer Institute*, **$100,000**
- **2024** NIH Loan Repayment Program (LRP) Award, *National Cancer Institute*, **$100,000**

- **2023** U54 (NIH Human Virome Program): Lead Analyst and architect of the Data Analysis & Submission Core for the center grant, enabling large-scale genomic data analysis and public data submission. This role involved designing reproducible computational infrastructure for harmonized virome data processing, metadata integration, and deposition across collaborating sites.
- **2023** Young Investigators Travel Award, *SMBE*, **$3,500**
- **2023** Opening Influenza Research Fellowship, *Center for Open Science*, **$2,000**
- **2022** CST Outstanding Research Award, *Temple University*, **$500**
- **Grant Applications**:
  - **NIH VAST Supplement:** Proposed development of scalable, AI-enabled frameworks for viral surveillance and genomic data integration supporting the NIH Human Virome Program, with emphasis on standardized bioinformatics workflows, benchmarking, and harmonized approaches for viral discovery and characterization across diverse sequencing datasets.
  - **Horizon Europe Grant:** Developed a multinational proposal to harmonize urban microbiome and virome data across global cities using FAIR, AI-ready data standards and predictive modeling. The proposal emphasized interoperable metadata, reproducible workflows, and comparative analyses across built environments.
  - **DOE Genesis Grant:** Designed a project to establish standardized, AI-ready multi-omics datasets and workflows for *Pseudomonas putida*, integrating proteomics, metabolomics, and lipidomics with FAIR-compliant metadata. The project framed microbial systems biology as a foundation for reusable, machine-readable datasets that support downstream modeling and discovery.
  - **Blueprint Biosecurity RFP:** Designed the proposal and technical strategy for an AI-enabled pathogen detection platform capable of identifying known, novel, zoonotic, and potentially engineered threats from metagenomic sequencing data. The project integrated scalable bioinformatics workflows, evolutionary and genomic inference, foundation-model approaches, and anomaly detection to support pathogen discovery, characterization, and biosecurity risk assessment. The proposal emphasized rigorous benchmarking across diverse microbial datasets, interpretable identification of previously uncharacterized sequences, and deployment-ready genomic surveillance tools for public health and biodefense applications.
- **Professional Membership**: Sigma Xi (The Scientific Research Honor Society)
- **Travel Awards**: SMBE (2020, 2023); EMBO (2023); Center for Open Science Metascience (2023)

SERVICE

- **Journal Reviewer**: *Microbiome* (2026); *Astrobiology* (2026); *Bioinformatics Advances* (2026, 2025); *Molecular Biology and Evolution* (2024); *Virus Evolution* (2023); *Genomics* (2021); ad hoc reviewer for journals in evolutionary genomics, virology, metagenomics, and computational biology.
  - Reviewed manuscripts in molecular evolution, microbial and viral genomics, metagenomics, genomic surveillance, and computational biology
  - Evaluated methodological rigor, reproducibility, statistical analysis, and biological interpretation
  - Recommended improvements to study design, validation, software, data availability, and presentation
- **Biological Science and Technology (BSAT) Study Group Member**,
  Defense Advanced Research Projects Agency (DARPA), 2025–2026
  - Contributed expert input on computational biology, genomics, biotechnology, and biosecurity

- Advised on emerging research directions at the intersection of biotechnology, national security, and data science
- Assessed scientific opportunities, technical bottlenecks, and potential applications of advanced genomic technologies
- Participated in interdisciplinary discussions on biological preparedness and emerging technology priorities

TEACHING EXPERIENCE

Weill Cornell Medicine: Quantitative Understanding in Biology Fall 2025
Guest Lecturer

· Delivered an invited module on metagenomics and metatranscriptomics within the course *Quantitative Information in Biological Systems*, integrating real-world urban microbiome case studies and demonstrating computational workflows for high-throughput sequencing data analysis.
· Emphasized quantitative reasoning, reproducible bioinformatics practices, and interpretation of multiomic datasets for students in quantitative biology and biomedical programs.

International Workshops and Summer Schools

· **Instructor, VEME 2023 & 2024 (International Workshop on Virus Evolution and Molecular Epidemiology)** Designed and delivered a half-day intensive module on molecular adaptation, including hands-on HyPhy-based selection analyses for an international audience of researchers in bioinformatics, virology, and public health.
· **Instructor, EEBG 2024 - 2026 (Eastern European Bioinformatics and Computational Genomics School)** Taught modules on molecular adaptation and metagenomics, combining conceptual lectures with practical training in evolutionary analysis workflows and metagenomic data interpretation.

Temple University Bioinformatics Studio Spring 2018 to 2023
Co-Founder

· Co-founded and scaled a student-led training initiative focused on accessible, hands-on bioinformatics education for learners ranging from beginners to advanced trainees.
· Developed and delivered workshops covering sequence analysis, metagenomics, and reproducible computational workflows.
· Mentored students and early-career researchers in computational biology best practices, fostering an inclusive peer-learning environment.

Temple University Genomics in Medicine (Graduate TA) Fall 2019 to 2021

· Supported enrollment of 100–160 students per semester in a cross-listed graduate/undergraduate genomics course.
· Led weekly office hours and review sessions, clarified primary literature, and provided detailed feedback on computational and written assignments.
· Assisted students with project design and interpretation of genomics data, strengthening quantitative and translational understanding.

Temple University Laboratory Instruction (Graduate TA) 2018 to 2019

· **Intro to Organismal Biology (Spring 2019)** Led two laboratory sections (40 students total), instructed experimental techniques, reinforced core biological concepts, and provided individualized feedback on lab performance.
· **General Biology I (Fall 2018)** Directed laboratory sessions for 40 students, demonstrated experimental protocols, and evaluated laboratory reports and practical skills.

New York University Adjunct Professor 2017 to 2018

- **Fundamentals of Bioinformatics (Spring 2018)** Led weekly recitations for 20 students, teaching core bioinformatics theory and software applications while supporting student projects.
- **Molecules of Life (Fall 2017)** Instructed two laboratory sections (40 students total), reinforcing lecture material through hands-on biochemical and molecular biology exercises.
- **Principles of Biology Lab (Spring 2017)** Taught two laboratory sections, guiding students through experimental design, data analysis, and foundational biological concepts.

PRESENTATIONS

Invited Talks

- **2026** American Museum of Natural History (SciCafe). *The Subway's Invisible Microbiome.* Presented how urban transit systems harbor diverse microbial communities that reflect human activity, geography, and environmental conditions. Discussed how metagenomic surveillance can reveal patterns of microbial diversity, antimicrobial resistance, and public-health relevance across cities.
- **2025** Clarkson University. *Challenges in Examining the Global Urban Microbiome.* Examined the technical, biological, and logistical challenges of comparing urban microbiomes across global sampling sites. Highlighted issues in study design, contamination control, data harmonization, reference databases, and interpretation of geographic and temporal variation.

Conference and Workshop Presentations

- **2026** ICCABS. Diversity and distinctive characteristics of the global RNA virome in urban & peri-urban environments.
- **2025** AGBT. Diversity and distinctive traits of the global RNA virome in urban environments.
- **2025** RdRp Summit. The global RNA virome in urban environments.
- **2025** CAMeRA. Challenges in examining the global urban microbiome.
- **2025** NASA GeneLab Microbes AWG. The global RNA virome in urban environments.
- **2024** MetaSUB Consortium. Quantifying natural selection in urban environments.
- **2023** Dynamics and Evolution. Selection analyses identify broad evolutionary features across viral families.
- **2023** Temple University. The role of complex evolutionary dynamics in molecular sequence analysis.
- **2022** Weill Cornell Medical College. Advances in quantifying natural selection in coding sequences.
- **2022** Dynamics and Evolution of Human Viruses. RASCL: Rapid assessment of selection in clades.
- **2021** EMBO. Widespread empirical support for instantaneous multiple-nucleotide changes.
- **2021** Mid-Atlantic Bioinformatics Conference. Rapid assessment of selection in SARS-CoV-2 variants.
- **2020** Cold Spring Harbor Laboratory. Widespread empirical support for instantaneous multiple-nucleotide changes.
- **2020** SMBE. Evaluating the impact of multiple simultaneous mutations on evolutionary rate inference.
- **2019** BGSS Retreat, Temple University. Evaluating the impact of multiple simultaneous mutations on evolutionary rate inference.